

Cox Model for The Risk of Relapse to Drug Abuse

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1 Introduction

The goal of this project is to develop a survival analysis model to understand the risk of relapse to drug abuse and identify which covariates significantly influence this risk.

2 Data description

2.1 Data

The dataset used in this project originates from a study conducted between 1989 and 1994 on the effect of preventive measures among drug addicts in Massachusetts, USA. The purpose of the study was to investigate whether different treatment programs could prevent further drug abuse. The data contains the following variables:

- **days**: Number of days from the start of treatment to relapse or censoring.
- **relapse**: Indicator for relapse to drug abuse (1) or censoring (0)
- **treat**: Planned duration of treatment (0 = three months, 1 = six months)
- **age**: age at start of treatment (in years)
- **race**: Race (0 = white, 1 = others)
- **num**: Number of earlier treatments for drug abuse
- **use**: Use of heroine/cocaine last three months (1 = both heroine and cocaine, 2 = only heroine, 3 = only cocaine, 4 = neither heroine nor cocaine)
- **dep**: Beck score for depression at start of treatment (high values of the score corresponds to a high degree of depression)

2.2 Explanatory Data Analysis

Figure 1 displays the structure of the dataset, which contains 575 observations and 8 variables.

	days	relapse	treat	age	race	num	use	dep
1	188	1	1	39	0	1	4	9
2	26	1	1	33	0	8	4	34
3	207	1	1	33	0	3	2	10
4	144	1	0	32	0	1	4	20
5	551	0	1	24	1	5	2	5
6	32	1	1	30	0	1	3	33

Figure 1: Structure of Data

Figure 2 provides a summary of the variables along with their histograms. The left panel displays descriptive statistics, including the minimum, maximum, median, and quartiles for each variable. The right panel shows histograms illustrating the distribution of each explanatory variable. The **dep** variable shows a relatively uniform distribution. The binary variables **relapse** and **treat** indicate that the data is balanced.

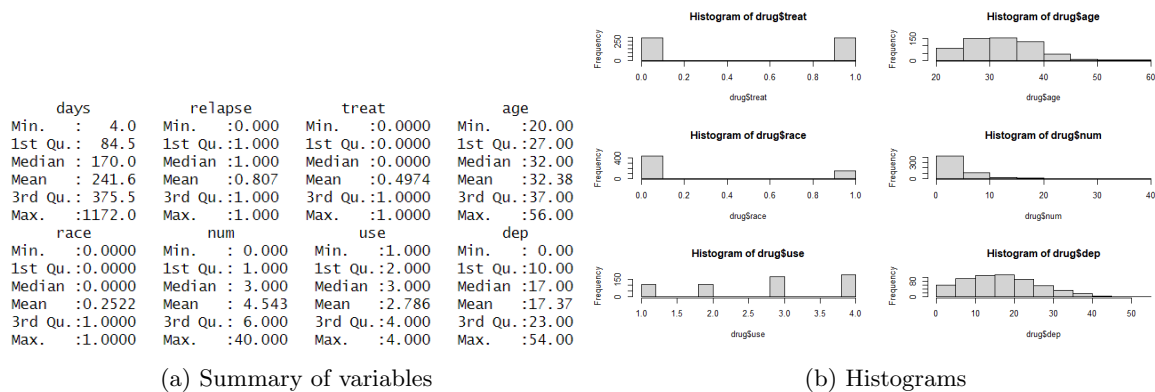
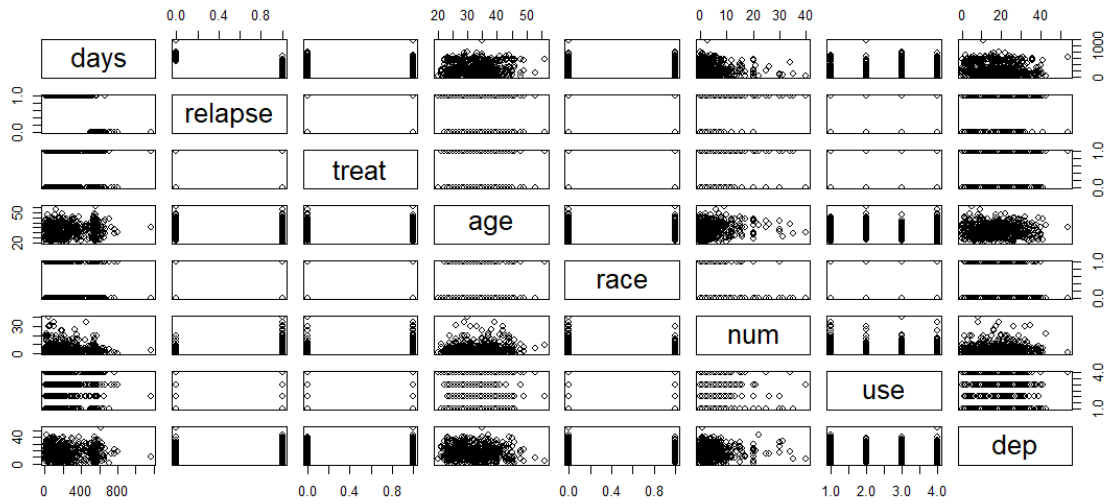


Figure 2

Correlation To investigate potential relationships between variables, correlation analysis was conducted, as shown in Figure 3. The correlation matrix reveals no strong correlations between explanatory variables and the response variables (**days** and **relapse**).



(a)

Parameter1	Parameter2	r	95% CI	t(573)	p
days	relapse	-0.82	[-0.84, -0.79]	-34.19	< .001***
days	treat	0.12	[0.04, 0.20]	2.85	0.096
days	age	0.07	[-0.01, 0.15]	1.71	> .999
days	race	0.11	[0.03, 0.19]	2.74	0.128
days	num	-0.13	[-0.21, -0.05]	-3.23	0.029*
days	use	0.08	[0.00, 0.16]	1.90	0.867
days	dep	-0.09	[-0.17, 0.00]	-2.08	0.653
relapse	treat	-0.05	[-0.13, 0.03]	-1.22	> .999
relapse	age	-0.06	[-0.14, 0.02]	-1.39	> .999
relapse	race	-0.10	[-0.18, -0.02]	-2.44	0.282
relapse	num	0.15	[0.06, 0.22]	3.51	0.012*
relapse	use	-0.07	[-0.15, 0.01]	-1.62	> .999
relapse	dep	0.06	[-0.02, 0.14]	1.43	> .999
treat	age	-0.04	[-0.13, 0.04]	-1.07	> .999
treat	race	0.08	[0.00, 0.16]	1.90	0.867
treat	num	-2.03e-03	[-0.08, 0.08]	-0.05	> .999
treat	use	0.08	[0.00, 0.16]	2.00	0.738
treat	dep	9.89e-03	[-0.07, 0.09]	0.24	> .999
age	race	0.01	[-0.07, 0.10]	0.33	> .999
age	num	0.20	[0.12, 0.27]	4.82	< .001***
age	use	-0.13	[-0.21, -0.05]	-3.24	0.029*
age	dep	-0.04	[-0.12, 0.04]	-0.89	> .999
race	num	-0.10	[-0.18, -0.02]	-2.37	0.325
race	use	0.05	[-0.03, 0.14]	1.32	> .999
race	dep	1.30e-03	[-0.08, 0.08]	0.03	> .999
num	use	-0.24	[-0.31, -0.16]	-5.83	< .001***
num	dep	0.06	[-0.02, 0.14]	1.41	> .999
use	dep	-0.15	[-0.22, -0.06]	-3.52	0.012*

p-value adjustment method: Holm (1979)
 Observations: 575

(b)

Figure 3: Correlation

KM fitting To explore the effect of each covariate on survival time, Kaplan-Meier (KM) survival curves were generated for both categorical and continuous variables. For continuous covariates, the data were divided into two groups based on the mean value. The KM curves for the covariates are presented in Figure 4.

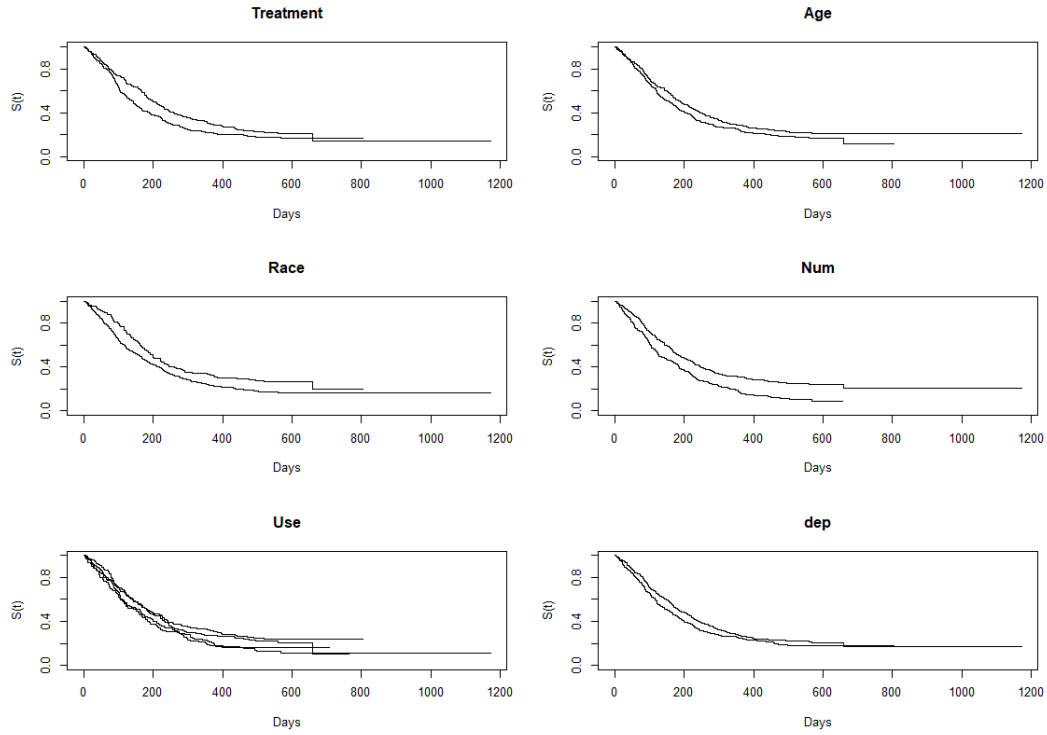


Figure 4: KM Plot for Each Variable

From these survival curves, we can observe that most covariates show noticeable differences between groups, indicating their potential importance in predicting survival time. However, the `use` variable shows considerable overlap between the groups. This suggests that the `use` variable may not be as important in predicting survival time compared to other covariates.

3 Model

3.1 Model Selection

To identify the best model for predicting relapse risk, we analyzed the covariates using KM survival curves. The KM analysis suggested that the **use** variable may not be significant due to overlapping survival curves between groups.

Based on these insights, we applied stepwise selection methods (forward, backward, and bidirectional) to refine the model. The final model was chosen using the Akaike Information Criterion (AIC), balancing model fit and complexity while excluding less informative covariates like **use**.

Table 1 shows the candidate models and their corresponding AIC values.

Table 1: Results of AIC for Candidate Models from Stepwise Selection

Model	Covariates	AIC
step_both	treat + age + race + num + dep	5299.550
step_for	treat + age + race + num + use + dep	5299.770
step_back	treat + age + race + num + dep	5299.550

Based on the AIC results, both the **step_both** and **step_back** methods identified this model as optimal. And the model with covariates **treat**, **age**, **race**, **num**, and **dep** was selected, as it yielded the smallest AIC value of 5299.550.

The selected model can be expressed using the Cox proportional hazards formula:

$$h(t) = h_0(t) \exp(\beta_1 \text{treat} + \beta_2 \text{age} + \beta_3 \text{race} + \beta_4 \text{num} + \beta_5 \text{dep}) \quad (1)$$

where $h(t)$ is the hazard function, $h_0(t)$ is the baseline hazard, and $\beta_1, \beta_2, \beta_3, \beta_4, \beta_5$ are the coefficients associated with the covariates **treat**, **age**, **race**, **num**, and **dep**, respectively.

The cumulative hazard plot for selected model is illustrated in Figure 5.

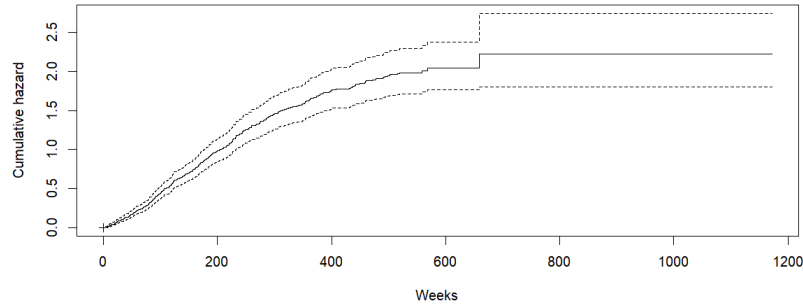


Figure 5: Cumulative Hazard plot for Selected Model

3.2 Diagnostic

3.2.1 For Selected Model

In order to validate the assumptions of the Cox proportional hazards model expressed in Equation (1), we checked for log-linearity and the proportional hazard (PH) assumption.

Log-linearity Figure 6 shows the results of assessing the log-linearity assumption by simulating 1000 values. The grey lines represent the simulation envelope under the log-linearity assumption, while the black line represents the relationship based on the observed data. For the `treat` covariate, the black line deviates from the grey envelope, indicating that the log-linearity assumption is violated for this covariate.

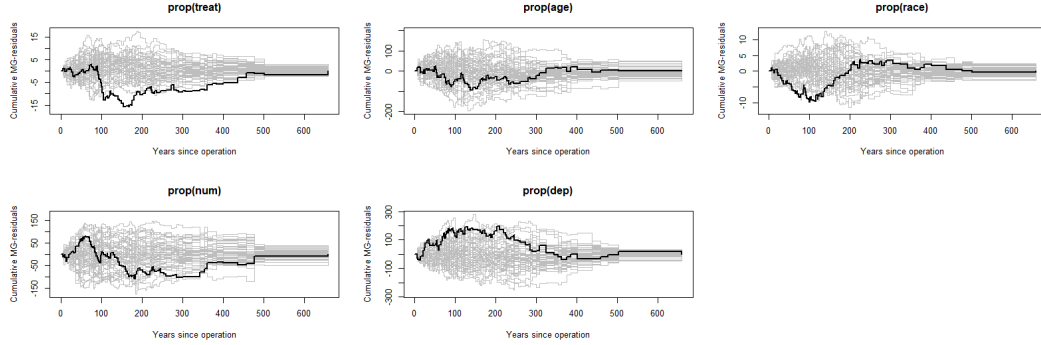


Figure 6: Log-linearity

Proportional Hazard Assumption To verify the proportional hazard assumption, the Schoenfeld residuals method was applied. Figure 7 illustrates the diagnostic results.

The left panel provide p-values of Schoenfeld test for each covariate. Since all p-values are greater than 0.05, we fail to reject the null hypothesis that the covariates satisfy the proportional hazard assumption.

In the right panel, the Schoenfeld residuals are evenly distributed around zero, providing additional evidence that the proportional hazard assumption is not violated.

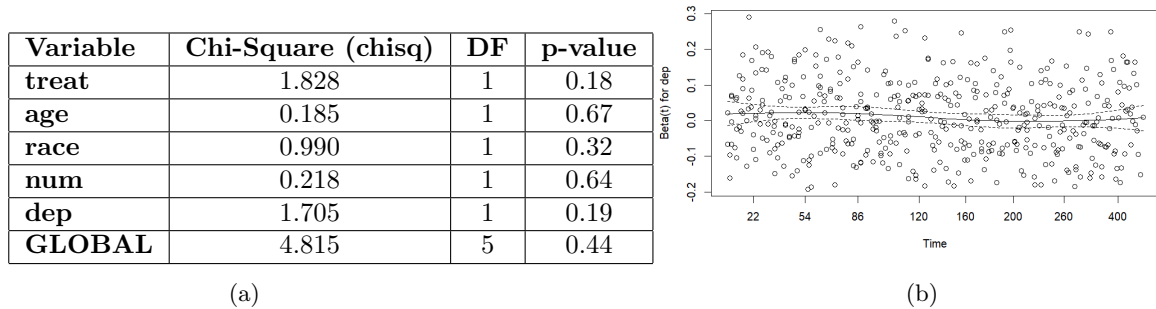


Figure 7: Schoenfeld Residual Test and Plot

Based on these diagnostics, the selected model meets the proportional hazard assumption, although the log-linearity assumption is violated for the `treat` covariate.

3.2.2 For Stratified Model

We refit the model by stratifying on the `treat` variable to account for differing baseline hazards between the treatment groups.

Log-linearity To validate the log-linearity assumption for the stratified model, diagnostic plots were generated and are shown in Figure 8. In the left panel, the black line remains within the grey envelope, so the log-linearity assumption is not violated for any of the covariates.

In the right panel, the plotted lines for all covariates appear linear, indicating that the log-linearity assumption holds for the stratified model.

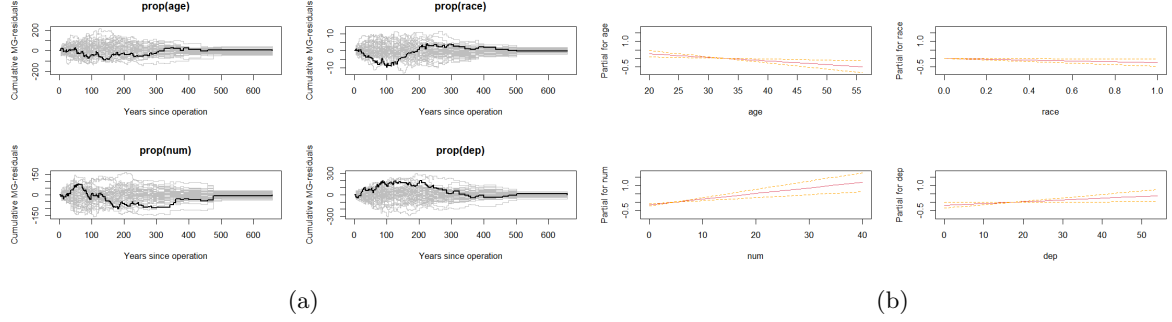


Figure 8: Log-linearity

Proportional Hazard Assumption To check the proportional hazard assumption, the Schoenfeld residuals test was performed, and the results are presented in Figure 9. In the left panel, all p-values exceed 0.05, indicating that none of the covariates violate the PH assumption. In the right panel, the residuals are evenly distributed around zero, providing additional evidence that the PH assumption is satisfied.

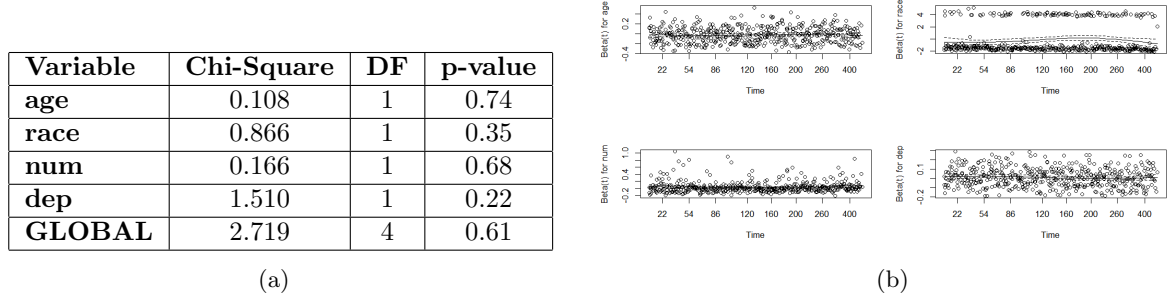


Figure 9: Schoenfeld Residual Test and Plot

Stratified Model Figure 10 shows the log cumulative hazard plot for the stratified model. The log cumulative hazard curves for the two treatment groups run nearly parallel, suggesting that the baseline hazards differ between the groups by a constant factor. Thus it provide validity of the stratified cox model.

In conclusion, the stratified model satisfies both the log-linearity and proportional hazard assumptions. The diagnostic checks confirm that the data fits well within the assumptions of the stratified Cox model for risk of relapse.

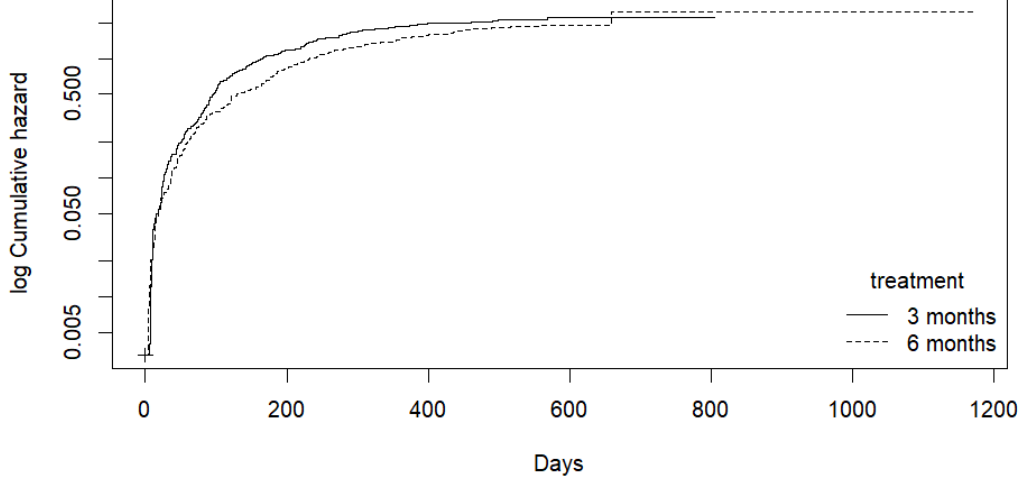


Figure 10: Log Cumulative Hazard Plot

3.3 Result

Our final model is a stratified Cox proportional hazards model, where the stratification is based on the `treat` variable. This allows the baseline hazard to differ between the treatment groups. The model can be expressed as follows:

$$h_i(t) = h_{i0}(t) \exp(\beta_1 \text{age} + \beta_2 \text{race} + \beta_3 \text{num} + \beta_4 \text{dep}), \quad (i=0: 3 \text{ months}, i=1: 6 \text{ months}) \quad (2)$$

where $h(t)$ is the hazard function at time t , $h_{i0}(t)$ is the baseline hazard specific to each treatment group, and the coefficients $\beta_1, \beta_2, \beta_3$, and β_4 correspond to the covariates `age`, `race`, `num`, and `dep`, respectively. The model fitting results are presented in Table 2.

Table 2: Model Coefficients

Variable	coef	exp(coef)	se(coef)	z	Pr(> z)
age	-0.021325	0.978901	0.007693	-2.772	0.00557 **
race	-0.252967	0.776493	0.111317	-2.272	0.02306 *
num	0.033469	1.034035	0.007919	4.227	2.37e-05 ***
dep	0.010543	1.010598	0.004894	2.154	0.03122 *

The interpretation of the model is as follows:

- **age** for each one-year increase in age, the hazard decreases by approximately 2.1% ($1 - 0.9789$).
- **race** participants classified as non-white have a hazard that is 77.6% that of white participants.
- **num** each additional previous treatment increases the hazard by approximately 3.4%. Therefore, participants with more previous treatments have a higher risk of relapse.
- **dep** For each one-unit increase in the Beck Depression Score, the hazard increases by 1.06%.

Figure 11 shows the cumulative hazard plot for the final model.

In conclusion, the final Cox proportional hazards model, stratified by `treat`, adequately captures the risk of relapse, satisfying both the log-linearity and proportional hazard assumptions.

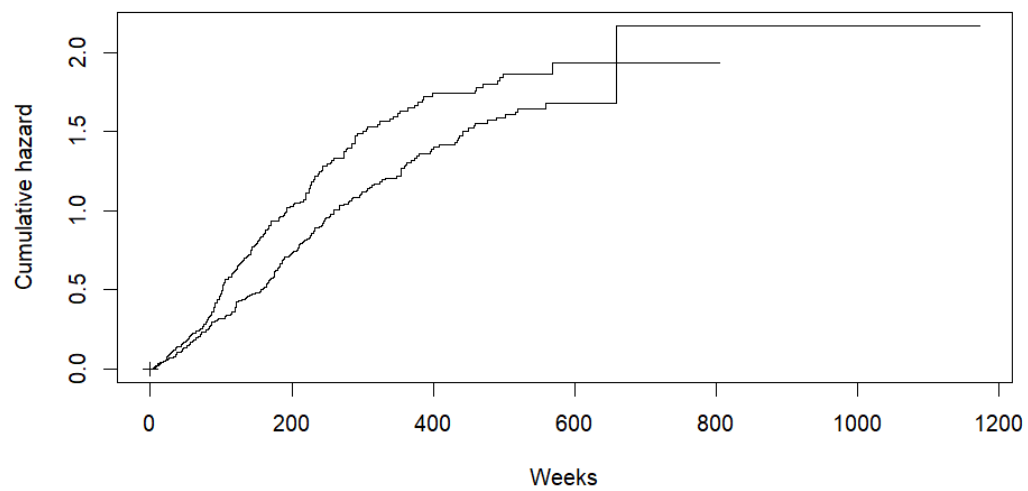


Figure 11: Cumulative Hazard Plot

4 Discussion and Conclusion

This project aimed to develop a Cox proportional hazards model to analyze the risk of relapse to drug abuse and identify significant covariates influencing this risk.

In the Explanatory Data Analysis phase, Kaplan-Meier (KM) survival curves were generated for each covariate to visually assess their impact on survival time. The analysis showed that most covariates, such as `treat`, `age`, `race`, `num`, and `dep`, had noticeable differences between groups. However, the `use` variable exhibited considerable overlap, indicating it may not be a significant predictor of relapse (Figure 4).

Based on these insights, we performed model selection using stepwise methods (forward, backward, and bidirectional) and evaluated the models using the Akaike Information Criterion (AIC). The models `step.both` and `step.back` yielded the smallest AIC values, leading to the selection of the final model, which included `trea`, `age`, `race`, `num`, and `dep` as covariates (Table ??).

Diagnostics were conducted to ensure the validity of the selected model. Upon assessing the log-linearity assumption, the initial model revealed a violation for `treat`. This issue was resolved by stratification based on `treat`. The stratified model satisfied both the log-linearity and proportional hazards assumptions (Figures 8 and 9). Additionally, the cumulative hazard plot showed that the log cumulative hazard curves for the two treatment groups were nearly parallel, supporting the validity of the stratified Cox model (Figure 10).

In conclusion, the final stratified Cox proportional hazards model, which included `age`, `race`, `num`, and `dep`, met the necessary model assumptions and provided a robust framework for understanding the factors influencing the risk of relapse to drug abuse.

5 Appendix

5.1 Code

```
1 library(survival)
2 library(timereg)
3
4 setwd("C:/Users/SUNA/Documents/R/survival")
5 drug <- read.table('drugs.txt', header = TRUE)
6
7 #EDA
8 head(drug)
9
10 summary(drug)
11
12 par(mfrow=c(3,2))
13 hist(drug$treat)
14 hist(drug$age)
15 hist(drug$race)
16 hist(drug$num)
17 hist(drug$use)
18 hist(drug$dep)
19
20 pairs(drug)
21 correlation(drug, method="pearson")
22
23 ##KM Curv
24 library(survminer)
25 par(mfrow=c(3,2))
26 km_fit1 <- survfit(Surv(days, relapse == 1) ~ treat, data = drug)
27 plot(km_fit1,main="Treatment", xlab="Days", ylab="S(t)")
28 drug$age.g<-as.numeric(drug$age>mean(drug$age))
29 km_fit2 <- survfit(Surv(days, relapse == 1) ~ age.g, data = drug)
30 plot(km_fit2,main="Age", xlab="Days", ylab="S(t)")
31 km_fit3 <- survfit(Surv(days, relapse == 1) ~ race, data = drug)
32 plot(km_fit3,main="Race", xlab="Days", ylab="S(t)")
33 drug$num.g<-as.numeric(drug$num>mean(drug$num))
34 km_fit4 <- survfit(Surv(days, relapse == 1) ~ num.g, data = drug)
35 plot(km_fit4,main="Num", xlab="Days", ylab="S(t)")
36 km_fit5 <- survfit(Surv(days, relapse == 1) ~ use, data = drug)
37 plot(km_fit5,main="Use", xlab="Days", ylab="S(t)")
```

```

38 drug$dep.g<-as.numeric(drug$dep>mean(drug$dep))
39 km_fit6 <- survfit(Surv(days, relapse == 1) ~ dep.g, data = drug)
40 plot(km_fit6,main=("dep"), xlab="Days", ylab="S(t)")

```

Listing 1: EDA

```

1 # Modeling
2 # stepwise
3 full_model <- coxph(Surv(days, relapse) ~ treat + age + race + num + use + dep, data =
  drug)
4 step_both <- step(full_model, direction = "both") #treat + age + race + num + dep
5 step_for<-step(full_model, direction = "for") #treat + age + race + num + use + dep
6 step_back<-step(full_model, direction = "back") #treat + age + race + num + dep
7
8 AIC( step_both,step_for, step_back)
9 select=step_both # explanatory: treat, age, race, num, dep
10
11
12 #res
13 surv <- survfit(select)
14 par(mfrow=c(1,1))
15 plot(surv, fun = "cumhaz", mark.time = F,
16       xlab = "Weeks", ylab = "Cumulative_hazard")

```

Listing 2: Model Selection

```

1 #Diagnostic
2 ## for select model
3 ### log linearity
4 fit.ut = cox.aalen(Surv(days, relapse == 1) ~ prop(treat)+prop(age)+prop(race)+prop(
  num)+prop(dep),
5                   data = drug, weighted.test = 0, residuals = 1,
6                   rate.sim = 0, n.sim = 1000)
7 par(mfrow = c(2, 3))
8 plot(fit.ut, score = T, xlab = "Years_since_operation")
9 ### ph assumption
10 cox.zph(select, transform = 'log')
11 par(mfrow = c(1, 1))
12 plot(cox.zph(select))
13
14 ## for statified model
15 fit.ut = cox.aalen(Surv(days, relapse == 1) ~ strata(treat)+prop(age)+prop(race)+prop
  (num)+prop(dep),
16 data = drug, weighted.test = 0, residuals = 1,
17 rate.sim = 0, n.sim = 1000)
18 ### log linearity 1
19 par(mfrow = c(2, 2))
20 plot(fit.ut, score = T, xlab = "Years_since_operation")
21
22 ### log linearity 2
23 fit.st = coxph(Surv(days, relapse == 1) ~ strata(treat)+age+race+num+dep,
24               data = drug)
25 par(mfrow = c(2, 2))
26 termplot(fit.st, se = T) #dia aft2
27
28 ### proportional assumption
29
30 cox.zph(fit.st, transform = 'log') #aft 5
31 par(mfrow = c(2, 2))
32 plot(cox.zph(fit.st)) #aft 6
33
34
35 ### plot of stratified model
36 fit.st = coxph(Surv(days, relapse == 1) ~ strata(treat)+age+race+num+dep,
37               data = drug)
38 baseline.covar = data.frame(age = 1,race=1,num=1,dep=1)
39 surv.st = survfit(fit.st, newdata = baseline.covar)
40 par(mfrow=c(1,1))
41 plot(surv.st, fun = "cumhaz", log = T,
42       mark.time = F, lty = 1:2,
43       xlab = "Days", ylab = "log_Cumulative_hazard")

```

```
44 legend("bottomright", c("3_months", "6_months"), title = "treatment", lty=1:2, bty = "n")
```

Listing 3: Diagnostic

```
1 #Result
2 fit.st = coxph(Surv(days, relapse == 1) ~ strata(treat)+age+race+num+dep,
3               data = drug)
4 summary(fit.st)
5 surv <- survfit(fit.st)
6 par(mfrow=c(1,1))
7 plot(surv, fun = "cumhaz", mark.time = F,
8       xlab = "Weeks", ylab = "Cumulative_hazard")
```

Listing 4: Resulting Model